Technical Service Bulletin

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Exhaust Gas Recirculation (EGR) Cooler Leaks Caused by Truck Wash Chemicals: Fault Code 197 and Internal Coolant Leak	

Exhaust Gas Recirculation (EGR) Cooler Leaks Caused by Truck Wash Chemicals: Fault Code 197 and Internal Coolant Leak

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- ISX12 G CM2180 EJ
- ISX12N CM2380 X120B
- X12 CM2350 X119B
- X12 CM2450 X137B
- X15 CM2350 X114B
- X15 CM2350 X116B
- X15 CM2450 X124B
- X15 CM2450 X134B
- X15 CM2450 X142B

Issue Summary

EGR cooler leaks coolant internally caused by corrosion of the individual gas tubes within the EGR cooler.

Symptom:

- Fault Code 197
- Internal coolant leak

Root Cause:

- Chlorides and fluorides in truck and trailer wash chemicals are being introduced to the EGR cooler through the engine air intake

Verification

- Verify fluid in EGR cooler is coolant and not condensation by using Fleetguard® 3-Way Coolant Test Kit, Part Number CC2602B. Follow the instructions on the coolant test strip package.
- An alternative to testing for coolant would be to pressure test the EGR cooler. See corresponding Service Manual. Reference Procedure 011-019 in Section 11 for EGR Cooler Pressure Test instructions.
- Review wash chemical(s) SDS to determine if hydrofluoric acid or other chemicals containing chlorine are used.
- Determine level of chloride parts per million (ppm) in the wash chemical(s) concentrate form. Test strips (such as Hach, Part Number 2744940) are available to test wash chemicals for high chloride levels. Contact 1-800-CUMMINS™ if help is needed determining chloride ppm level in the wash chemical(s).

Resolution

- Do **not** use wash chemicals that contain hydrofluoric acid. Wash chemicals free of hydrofluoric acid are available.
- Source an equivalent wash chemical with a lower chloride level. Chloride levels can range from several hundred ppm to thousands of ppm.
- Prevent wash chemicals from entering the engine air intake system. Cover the engine air intake during the wash process. See Figure 1.
- Do **not** operate the engine during the wash process.
- Do **not** park the unit for longer than 1 hour immediately after washing. Shutting the engine off immediately after washing allows these chemicals to condense in the EGR cooler and initiate the corrosion process.

Figure 1, Bucket Covering the Engine Air Intake During the Wash Process.

Document History

Date	Details
2022-6-9	Module Created
2024-2-29	Updated Product Affected, statement on test strips, and Figure 1.

Last Modified: 29-Feb-2024